

Module 3 Lesson 2

Graphing and Functions (7.1)

Objectives

In this lesson, we will learn how to

- plot points in the rectangular coordinate system;
- graph equations in the rectangular coordinate system;
- use function notation;
- graph functions;
- use the vertical line test;
- obtain information about a function from its graph.

1. Points and Ordered Pairs

The rectangular coordinate system is formed by two number lines that intersect at right angles.

Coordinate Axes

- The horizontal number line is the **x-axis**.
- The vertical number line is the **y-axis**.
- The point where the axes intersect is called the **origin**.

The origin is written as

$(0, 0)$.

Negative numbers are shown to the left of the origin on the x -axis and below the origin on the y -axis.

The axes divide the plane into four regions called **quadrants**.

Ordered Pair

Each point in the rectangular coordinate system corresponds to an ordered pair

$$(x, y).$$

The first number, x , is called the **x-coordinate**.

The second number, y , is called the **y-coordinate**.

The x -coordinate tells us how far to move left or right from the origin.

The y -coordinate tells us how far to move up or down from the origin.

2. Plotting Points

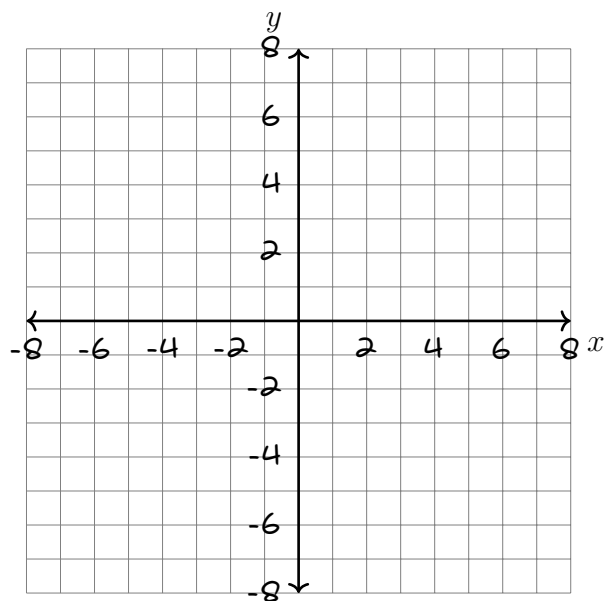
To plot a point (x, y) :

1. Start at the origin.
2. Move left or right according to the x -coordinate.
3. Move up or down according to the y -coordinate.
4. Mark the point.

Example 1: Plot the following points in the rectangular coordinate system:

$$A(-3, 5), \quad B(2, -4), \quad C(5, 0),$$

$$D(-5, -3), \quad E(0, 4), \quad F(0, 0).$$



Solution

Move from the origin as follows:

$A(-3, 5)$: 3 units left and 5 units up.

$B(2, -4)$: 2 units right and 4 units down.

$C(5, 0)$: 5 units right and 0 units up or down.

$D(-5, -3)$: 5 units left and 3 units down.

$E(0, 4)$: 0 units left or right and 4 units up.

$F(0, 0)$: the origin.

3. Graphs of Equations

A relationship between two quantities can be expressed as an equation in two variables.

For example,

$$y = 4 - x^2.$$

A **solution** of an equation in two variables is an ordered pair (x, y) that makes the equation true.

Graph of an Equation

The graph of an equation in two variables is the set of all points whose coordinates satisfy the equation.

4. Graphing by Point Plotting

To graph an equation by the point-plotting method:

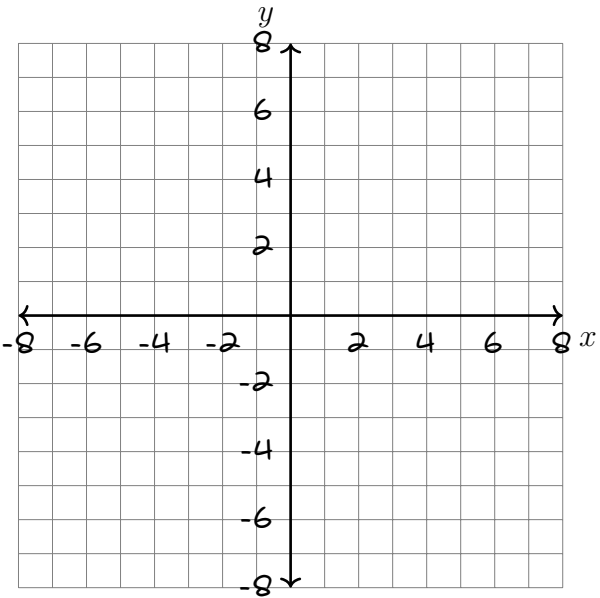
1. Choose values for x .
2. Substitute each value into the equation to find y .
3. Write the ordered pairs.
4. Plot the points.
5. Connect the points with a smooth curve or line when appropriate.

Example 2: Graph the equation

$$y = 4 - x^2.$$

Select integer values of x from -3 to 3 .

x	$y = 4 - x^2$	(x, y)
-3		
-2		
-1		
0		
1		
2		
3		



Solution

Compute the values:

x	$y = 4 - x^2$	(x, y)
-3	$4 - 9 = -5$	$(-3, -5)$
-2	$4 - 4 = 0$	$(-2, 0)$
-1	$4 - 1 = 3$	$(-1, 3)$
0	$4 - 0 = 4$	$(0, 4)$
1	$4 - 1 = 3$	$(1, 3)$
2	$4 - 4 = 0$	$(2, 0)$
3	$4 - 9 = -5$	$(3, -5)$

Plot the points and connect them with a smooth curve.

5. Functions

An equation in two variables may or may not describe a function.

Function

If an equation in two variables x and y gives exactly one value of y for each value of x , then y is a function of x .

The notation

$$y = f(x)$$

means that y is a function of x .

The expression $f(x)$ is read as "f of x."

Function Notation

If

$$f(x) = 2x + 4,$$

then $f(3)$ means the value of the function when $x = 3$.

Example 3: For

$$f(x) = 2x + 4,$$

find each value:

$$f(0)$$

$$f(3)$$

$$f(-2)$$

Solution

$$f(0) = 2(0) + 4 = 4.$$

$$f(3) = 2(3) + 4 = 10.$$

$$f(-2) = 2(-2) + 4 = 0.$$

6. Application Involving Function Notation

Functions are often used to model real-world situations.

For example, a function can model a car's required stopping distance based on speed.

Example 4: Suppose a function $f(x)$ models a car's required stopping distance, in feet, on dry pavement when the car is traveling x miles per hour.

Use the graph or model to estimate the required stopping distance at

60 miles per hour.

Round to the nearest 10 feet if using a graph.

Solution

From the graph, the required stopping distance at 60 miles per hour is approximately

360 feet.

Using the model, the required stopping distance is approximately

358 feet.

7. Graphing Functions

To graph a function, we can make a table of values and plot the ordered pairs.

Example 5: Graph the functions

$$f(x) = 2x$$

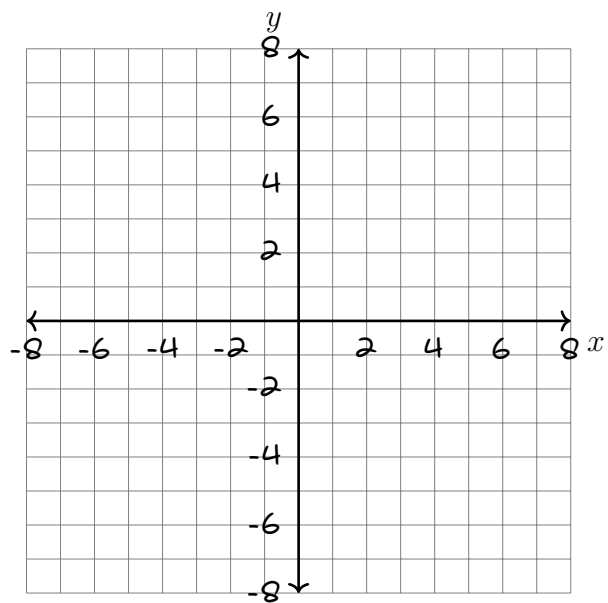
and

$$g(x) = 2x + 4$$

in the same rectangular coordinate system.
Select integers for x from -2 to 2 , inclusive.

x	$f(x) = 2x$	$(x, f(x))$
-2		
-1		
0		
1		
2		

x	$g(x) = 2x + 4$	$(x, g(x))$
-2		
-1		
0		
1		
2		



Solution

For $f(x) = 2x$:

x	$f(x) = 2x$	$(x, f(x))$
-2	-4	$(-2, -4)$
-1	-2	$(-1, -2)$
0	0	$(0, 0)$
1	2	$(1, 2)$
2	4	$(2, 4)$

For $g(x) = 2x + 4$:

x	$g(x) = 2x + 4$	$(x, g(x))$
-2	0	$(-2, 0)$
-1	2	$(-1, 2)$
0	4	$(0, 4)$
1	6	$(1, 6)$
2	8	$(2, 8)$

The graph of $g(x) = 2x + 4$ is the graph of $f(x) = 2x$ shifted up 4 units.

8. Vertical Line Test

The vertical line test helps determine whether a graph represents y as a function of x .

Vertical Line Test

If any vertical line intersects a graph in more than one point, then the graph does not define y as a function of x .

If every vertical line intersects the graph at most once, then the graph defines y as a function of x .

Example 6: Use the vertical line test to decide whether each graph represents y as a function of x .

- a. A circle
- b. A straight line that is not vertical
- c. A parabola opening upward
- d. A sideways parabola

Solution

- a. A circle is not a function because some vertical lines intersect it twice.
- b. A nonvertical straight line is a function.
- c. A parabola opening upward is a function.
- d. A sideways parabola is not a function because some vertical lines intersect it twice.

9. Analyzing the Graph of a Function

Graphs can give useful information about real-world situations.

For example, a graph may show body temperature over time.

Example 7: The graph illustrates body temperature from 8 a.m. through 3 p.m.

Let x be the number of hours after 8 a.m., and let y be the body temperature at time x .

Answer the following questions:

- What is the temperature at 8 a.m.?
- During which period of time is the temperature decreasing?
- Estimate the minimum temperature during the time period shown. When does it occur?
- During which period of time is the temperature increasing?
- Part of the graph is shown as a horizontal line segment. What does this mean?
- Explain why the graph defines y as a function of x .

Solution

- a. At 8 a.m., $x = 0$. The temperature is approximately

101°F.

- b. The temperature is decreasing when the graph falls from left to right. This occurs from $x = 0$ to $x = 3$, or from 8 a.m. to 11 a.m.

- c. The minimum temperature is approximately

98.6°F.

It occurs at $x = 3$, which is 11 a.m.

- d. The temperature is increasing from $x = 3$ to $x = 5$, or from 11 a.m. to 1 p.m.

- e. A horizontal line segment means the temperature is constant. The temperature stays at about

100°F

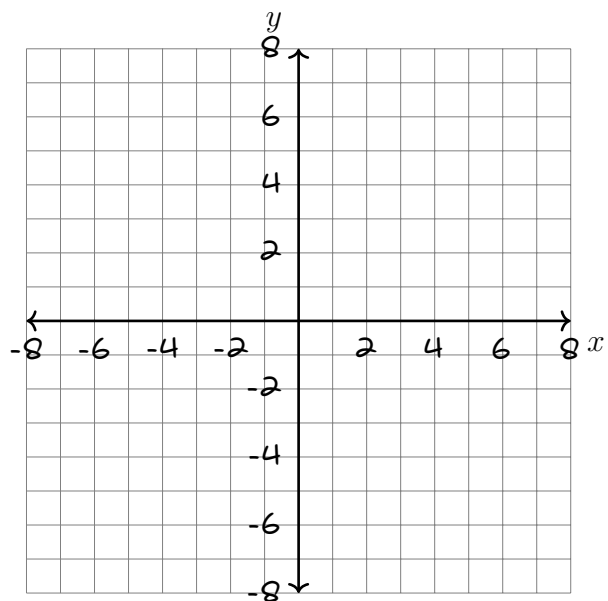
from $x = 5$ to $x = 7$, or from 1 p.m. to 3 p.m.

- f. The graph defines y as a function of x because each time corresponds to exactly one body temperature.

10. More Practice

Example 8: Plot the points:

$A(4, 3)$, $B(-2, 1)$, $C(-4, -5)$, $D(0, -3)$.



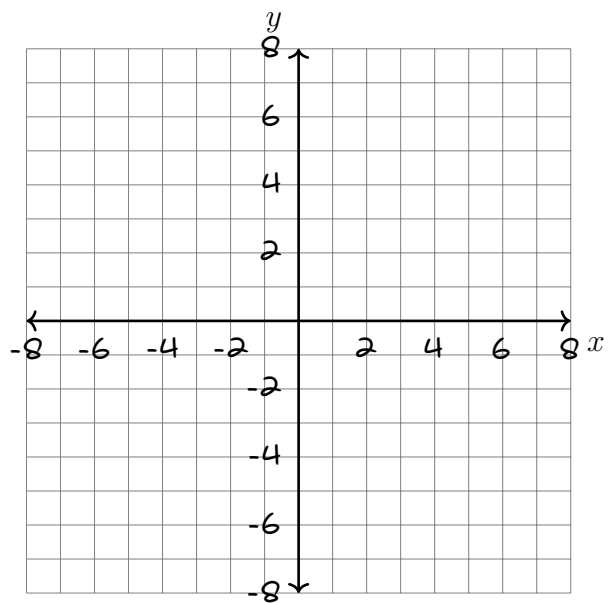
Example 9: Graph the equation

$$y = x^2 - 2$$

using the x -values

$-2, -1, 0, 1, 2$.

x	$y = x^2 - 2$	(x, y)
-2		
-1		
0		
1		
2		



Example 10: For

$$h(x) = x^2 + 3,$$

find

$$h(0), \quad h(2), \quad h(-3).$$

Summary

Key Ideas

- The rectangular coordinate system uses the x -axis and y -axis.
- An ordered pair (x, y) gives the location of a point.
- The x -coordinate tells us how far to move left or right.
- The y -coordinate tells us how far to move up or down.
- The graph of an equation is the set of all points that satisfy the equation.
- A function gives exactly one output value for each input value.
- Function notation $f(x)$ is read “ f of x .”
- The vertical line test helps determine whether a graph represents a function.
- Graphs can be used to obtain real-world information.